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Words by [Miriam Pinski](#)

The everything theory of mass transit

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8 Mins 

Who benefits when we ask transit to pursue social goals?

We ask public transport to address problems it was never designed to solve. Bikeshare services are charged with reducing emissions and improving local air quality. Subway stations need to beautify the neighborhood. Buses are made to add union jobs. And every new transit project should reflect community input and empower

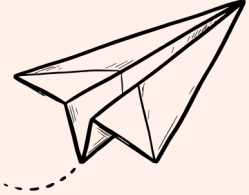


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residents. All of this impedes public transport’s ability to do its main job: transporting the public.

All these goals have merit. But each costs money and adds delay. The [Transit Costs Project](#) documents how the Boston Green Line extension, completed a decade later than anticipated, added a three-kilometer bike and pedestrian path after 13 people advocated for it at a community meeting, at a staggering cost of \$20 million.



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Buses are a rounding error as a proportion of vehicles on the road, accounting for less than one percent of US transportation emissions. Yet they operate under more rules and regulations than personal cars do. Governments build roads, highways, and parking structures with much more focus on their primary mandate: to improve mobility. But transit agencies answer to a far longer list of requirements.

Transit side quests

Transit missions didn’t always creep. Until the middle of the twentieth century, almost all transit was privately owned, and the goals were simple: maximize ridership and fare revenue while minimizing costs. But starting in the 1910s, transit companies struggled because of competition from the car.

In the 1960s, governments began taking over languishing transit lines. This gave rise to a new problem: transit came to rely on taxpayer subsidies, with the majority of funding coming from voters who didn’t use it. Most of the public now drove, so the case for continued subsidy had to be made based on non-users’ interests. Public transit aligned itself with the growing social movements of the 1960s and 1970s: the War on Poverty, the Civil Rights Movement, environmentalism, and the oil crisis. Today, the [American Public Transit Association](#) touts public transportation as a cure-all: transit investments, it says, ‘will get Americans to work, to education, and to healthcare, and it will also get our nation to more job creation, less traffic congestion and cleaner air’.

The typical transit agency gets funding from federal, state, and local sources. Each level of government can add more goals. The federal government might require domestically sourced materials (such as Buy America), daycare for manufacturers (like the [CHIPS Act](#)), or a higher minimum wage for contractors and

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The transit tradeoff

Alongside my co-authors, [I studied six transportation pilot programs](#) that were tasked with various goals, among them: reducing emissions, improving mobility equity, and reducing miles driven in cars. All of the pilots, which included electric carshare, bikeshare, and microtransit services, were funded by the California Air Resources Board and were located in [disadvantaged](#) areas across California.

By and large, these programs did improve the mobility of the badly off: most users lived in households earning under \$25,000 a year, and many of the trips were necessary to get to work, school, or medical appointments. But the programs added 128,880 net new vehicle miles and slightly increased overall emissions rather than decreasing them. As mobility schemes, they worked, but on other margins they did not.

The problem is not just specific equity or environmental programs. Existing services are being eroded by attempts to make buses electric. Battery electric buses cost more to procure, and parts are more expensive to replace. Austin’s CapMetro 40-foot electric buses cost \$917,000 apiece, whereas their diesel buses cost just \$535,000. In south Florida, the Miami-Dade county’s transit agency spent approximately \$1.1 million per electric bus. On any given day, out of the 75 buses it purchased, only about six are operating. That’s because the electric buses break down every 600 miles (seven times as often as their diesel equivalents), and because parts are difficult to come by, particularly since the manufacturer, Proterra, filed for bankruptcy.

In Germany, electrifying bus fleets increased costs by 12 percent. The problem is even worse in the United States because of Buy American rules. In Europe, where Chinese manufacturers compete freely, a standard 12-meter electric bus sells for roughly \$600,000. US agencies pay nearly twice as much. While demand for electric

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connections for charging infrastructure, and worker time to fuel vehicles. Just routing electric buses around charging infrastructure adds significant complexity to operating the service.

Every dollar spent complying with Buy America or staging community meetings is a dollar not spent improving frequency or coverage. Assuming the median cost per hour for operating a bus in the US is \$131, and that the median US bus carries 14.3 passengers per hour, the money for electrifying one bus could fund roughly 2,300 additional service hours, carrying at least 30,000 additional riders.

Prices make perfect

Electrifying transit is one way to reduce emissions, but it’s not the most efficient. Researchers studying Stockholm’s busiest bus route found that better pricing roads and transit, and running buses more

often, yielded more benefit than just electrifying buses. From a passenger perspective, electric buses are no different. But operators lose out because costs rise by at least 16 percent, depending on the charging setup. Combining road charges with more frequent buses entices drivers to shift to transit, reducing emissions more substantially than electrifying a few buses.

When transit becomes more viable and reliable, we see a positive feedback loop. People who would otherwise drive opt for transit, which boosts revenue and feeds back into improved quality. On average, increasing transit service frequency by 1 percent boosts ridership by about 0.5 percent. Once buses arrive every ten minutes, bus services become more appealing for riders with alternatives. These switches are the only kind that can make a substantial dent in emissions.

Congestion pricing would reduce emissions much more effectively than electrifying buses. In the few places brave enough to try, the ~~outcomes have been significant. In London, greenhouse gas~~

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where people and destinations cluster together. Switching people from driving to transit saved nine million metric tons of carbon in 2018. But we can prevent 66 million metric tons of carbon emissions by decreasing the distances people need to travel through more efficient land use. When distances are short, walking, biking, and transit all become more appealing, and when people do drive, they drive shorter distances. In major cities, where transit is more dominant, electrifying buses makes more sense than it does in car-dominant places, like Fresno, California, where it’s more important to move low-income people to their jobs.

Focusing transit on what it does best

In most of the US, not having a car makes poverty worse. Poor people who gain access to a car are more likely to find jobs, keep their jobs, find higher paying jobs, work more hours, get off welfare, and move to better neighborhoods. Even beyond

economic outcomes, people who don’t have cars take part in fewer activities outside the home, have fewer extracurricular activities and schools to choose from, fewer grocery store options, and are more likely to delay medical care.

Scope creep fails on its own terms. We are asking people already struggling to make ends meet, who already emit far less than most Americans do, to save the planet. But this is self defeating. Every dollar we spend on the environment or union jobs is a dollar we don’t spend on making transit more competitive against cars.

A 2018 UCLA report on transit ridership in Southern California estimated that if just one in every four people who rarely rides transit replaced a single driving trip with transit every two weeks, annual ridership would grow by 96 million. It’s more sensible to reduce emissions by getting a broader base to make small behavioral changes than to force a small group of already burdened people to carry the load.

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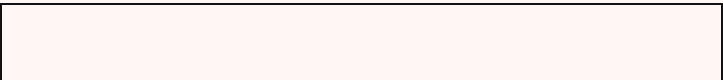
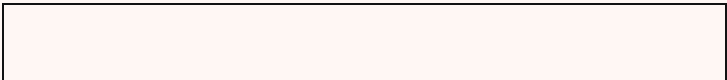


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